# Sustainable Fiber-based Electronic Systems Enabled by Highly Conductive, Deformation-Invariant Biodegradable Fiber Electrodes

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# 고전도성 및 변형 불변 특성을 갖는 생분해성 섬유 전극 기반 지속가능 섬유형 전자 시스템

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**Abstract**

Fiber-based electronic devices are promising platforms for wearable and implantable applications due to their flexibility, lightweight nature, and structural adaptability. The increasing use of electronic systems, however, has raised concerns regarding electronic waste, driving interest in biodegradable electronic technologies. Achieving stable and high-performance biodegradable fiber electrodes remains a key challenge, particularly under mechanical deformation. In this work, we develop a highly conductive, biodegradable fiber electrode with deformation-invariant properties for sustainable fiber-based electronic systems. The electrode is fabricated by dispersing zinc nanoparticles in a polypropylene carbonate (PPC) matrix via dip-coating and chemical sintering. The resulting conductive fiber exhibits low electrical resistance and maintains stable electrical performance under severe mechanical deformation. Its biodegradability is systematically evaluated under various environmental conditions.

Keywords: Sustainable fiber-based electronics system (지속가능한 섬유형 전자소자 시스템), Deformation-insensitive property (변형 무감응형 특성), Fiber electrode (섬유형 전극 소재), Biodegradable materials (생분해성 물질).

## 1. Introduction

Fibers have long been integral to textile manufacturing and daily life[1]. Their inherent softness and flexibility allow fiber-based garments to naturally conform to the body's curved contours, ensuring high comfort and wearability. With rapid advances in electronics, wearable and implantable devices for biosignal monitoring and energy harvesting have been actively developed. In this context, fiber-based electronic devices have gained increasing attention, as their fibrous form factor provides high flexibility, lightweight characteristics, and structural adaptability, making them well-suited for wearable and implantable applications[2]. Despite these advances, conventional electronic devices continue to generate substantial electronic waste, intensifying environmental pollution[3]. As a sustainable alternative, biodegradable electronic platforms that degrade under environmental stimuli, such as light, heat, and microorganisms, have been explored [4,5]. However, their broader adoption remains limited by complex fabrication processes, restricted material choices, and insufficient electrical conductivity and stability, particularly for high-performance fiber-based sensors.

In this study, we present highly conductive and deformation-invariant biodegradable fiber electrodes for sustainable fiber-based electronic systems. The electrode is fabricated by dip-coating zinc nanoparticles dispersed in a polypropylene carbonate (PPC) matrix, followed by chemical sintering. The resulting conductive fiber exhibits low electrical resistance (~0.7 Ω/cm) and excellent mechanical durability, maintaining stable electrical performance under severe mechanical deformation, including knotting, twisting, weight loading, and tape peeling. Furthermore, its biodegradability is systematically evaluated under various environmental conditions. Furthermore, the fabricated biodegradable fiber electrode was applied to an embroidery machine process to realize complex conductive patterns, including electrode structures designed for LED integration. In addition, EMG electrodes were fabricated using the proposed fiber electrode, and a wearable sleeve capable of measuring electromyographic signals under various body motions was successfully demonstrated.

## 2. Result and Discussion

The zinc/polypropylene carbonate (PPC)-based biodegradable fiber electrode was fabricated using a dip-coating process (Fig. 1a). The Zn/PPC conductive composite was deposited onto a PCL-coated PGA core fiber via dip-coating. Following the coating step, the fiber electrode was immersed in acetic acid to enhance its electrical conductivity (Fig. 1b). During this chemical sintering, the native oxide layer on the zinc nanoparticles was selectively dissolved, thereby promoting the formation of continuous electrical pathways between adjacent particles. As a result, the biodegradable fiber electrode exhibits excellent flexibility along with high electrical conductivity of approximately 0.6 Ω/cm (Fig. 1c and 1d).

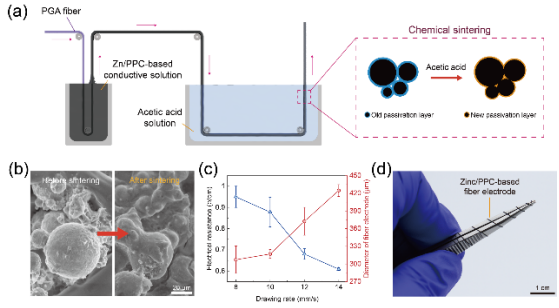


**Figure 1.** (a) Schematic illustration of the dip-coating process for the fabrication of a biodegradable fiber electrode. (b) SEM image of zinc nanoparticles in a fiber electrode. (c) Electrical resistance and cross-sectional area change with drawing speed. (d) Photograph of the biodegradable fiber electrode.

The fabricated biodegradable fiber electrode exhibited not only high electrical conductivity but also excellent electrical stability. When the electrical resistance of the fiber electrode was evaluated under bending deformation, only negligible changes of less than 1% were observed (Fig. 2a, b). Furthermore, the electrical resistance remained stable without performance degradation even after 10,000 repeated bending cycles, demonstrating the outstanding mechanical durability of the fiber electrode (Fig. 2c). In addition, when the fabricated fiber electrode was immersed in a lipase-based solution that simulates the in vivo environment, the Zn/PPC-based fiber electrode gradually dissolved over time, confirming its biodegradability (Fig. 2d, e).

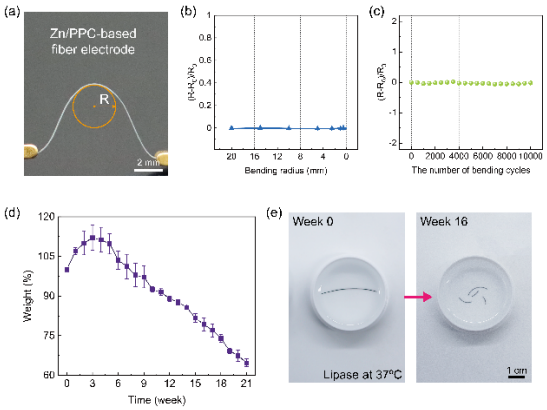


**Figure 2.** (a) Photograph of the biodegradable fiber electrode under bending deformation. (b) Electrical resistance variation of the fiber electrode as a function of bending radius. (c) Durability test of the fiber electrode under 10,000 repeated bending cycles. (d, e) Evaluation of the biodegradation behavior of the biodegradable fiber electrode.

Based on the mechanical stability and durability of the fiber electrode under deformation, the fabricated Zn/PPC-based biodegradable fiber electrode was applied to an embroidery machine process. Owing to its excellent electrical performance, the embroidered biodegradable fiber electrode exhibited minimal degradation in electrical properties even after the embroidery process, while enabling the facile fabrication of complex patterns (Fig. 3a, b).

Furthermore, the fiber electrode was patterned through embroidery to form conductive structures connected to light-emitting diodes (LEDs). Multiple LEDs were successfully integrated and illuminated, confirming stable electrical performance even under mechanical deformation (Fig. 3c). These results demonstrate the potential of the proposed fiber electrode for use in a wide range of future wearable and implantable devices, including EMG sensors and strain/pressure sensing systems.

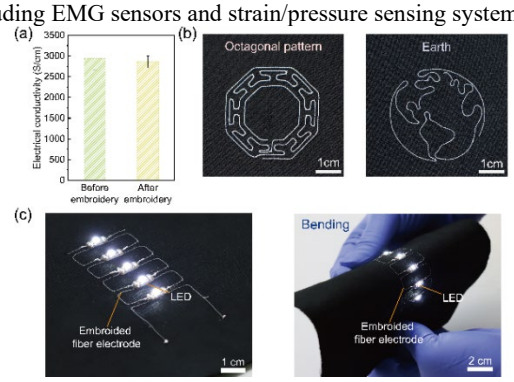


**Figure 3.** (a) Electrical conductivity change of the fiber electrode before and after the embroidery process. (b) Optical images of patterned fiber electrodes fabricated via the embroidery process. (c) Demonstration of an embroidery-based LED pattern and evaluation of its electrical performance under mechanical deformation.

## Conclusion

In conclusion, we developed a Zn/PPC-based biodegradable fiber electrode with deformation-insensitive electrical behavior and excellent mechanical durability. The fiber electrode maintained stable electrical conductivity under severe mechanical deformation and repeated bending, while exhibiting controllable biodegradability in biologically relevant environments. Moreover, its compatibility with embroidery processes enabled the fabrication of complex conductive patterns and the successful integration of LEDs without performance degradation. These results demonstrate the potential of the proposed fiber electrode as a sustainable platform for wearable and implantable electronics, including biosignal monitoring and strain or pressure sensing applications.

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